Draft - expert review pending

Note: This unit is complete and passes the platform's automated checks, but it is not reviewed yet. So, treat its sources and claims with care. You are encouraged to report errors and provide feedback.

Unit 10 assessment

Unit summary

This unit, which follows Chapter 10 of the course guide, moved from the probability of a single event to the probabilities of all the values of a number decided by chance.

- Topic 10.1 defined a random variable as a rule that gives a number to every outcome of a chance process. Discrete random variables come from counting and take separate values with gaps between them, such as the number of absentees. Continuous random variables come from measuring and can take any value in an interval, such as height or time. Rounding a measurement when we record it does not make it discrete.
- Topic 10.2 described a discrete random variable by its probability distribution, a list of every value with its probability, built from equally likely outcomes or from records. A valid distribution passes two checks: every probability lies between 0 and 1, and the probabilities add to 1. The expected value, the sum of each value times its probability, is the long-run average and need not be the most likely value, or even a possible one. A continuous random variable can be represented by a density curve instead, and the probability of an interval is the area under the curve.

Summative assessment

Multiple-choice questions (MCQs)

Choose the single best answer for each question.

1. *(Remember)* A random variable is:
   - A) A rule that gives a number to each outcome of a chance process
   - B) Any letter used for an unknown in algebra
   - C) A number that is the same every time the process is repeated
   - D) A list of outcomes written in words

2. *(Understand)* Which of these is a discrete random variable?
   - A) The time a pupil takes to run 100 metres
   - B) The number of pupils wearing glasses in a class
   - C) The weight of a pupil chosen at random
   - D) The volume of milk in a carton

3. *(Understand)* Which of these is a continuous random variable?
   - A) The number of brothers a pupil has
   - B) The number of correct answers on a quiz
   - C) The length of a pupil's arm
   - D) The number of buses arriving at a stop in an hour

4. *(Apply)* Three coins are tossed and X is the number of heads. Which values can X take?
   - A) 1, 2, 3
   - B) 0, 1, 2
   - C) 0, 1, 2, 3
   - D) 0 to 6

5. *(Apply)* A school chooses a pupil at random. Consider the pupil's actual age since birth, before rounding. This random variable is:

- A) Discrete, because it is recorded in whole numbers
- B) Continuous, because age can take any value between whole years
- C) Discrete, because ages are counted
- D) Neither, because age does not depend on chance

6. *(Remember)* In a valid probability distribution, the probabilities must:

   - A) All be equal
   - B) Each lie between 0 and 1 and add up to 1
   - C) Add up to 100
   - D) Increase as the value of x increases

7. *(Apply)* A distribution has $P(X = 0) = 0.2$, $P(X = 1) = 0.3$, $P(X = 3) = 0.1$, and X can take no other value except 2. What must $P(X = 2)$ be?

   - A) 0.3
   - B) 0.4
   - C) 0.5
   - D) 0.6

8. *(Apply)* X takes the values 0, 1 and 2 with probabilities 0.5, 0.3 and 0.2. What is the expected value of X?

   - A) 0.7
   - B) 1.0
   - C) 1.2
   - D) 0.33

9. *(Understand)* The expected number of absentees in a class is 2.12 per day. This means that:

   - A) 2.12 pupils will be absent tomorrow
   - B) Over many days, the average number absent will be about 2.12

- C) Exactly 2 absentees is impossible
- D) Most days have between 2.1 and 2.2 absentees

10. *(Understand)* For a continuous random variable, the probability that its value falls in an interval is shown by:
    - A) The height of the curve at one point
    - B) The area under the curve over that interval
    - C) The number of values inside the interval
    - D) Half the width of the interval

Restricted-response questions (RRQs)

Answer each question in a few sentences or a short paragraph.

1. *(Understand)* Using "the number of heads when two coins are tossed", explain in two sentences what makes a variable a *random* variable.

2. *(Apply)* Two coins are tossed and X is the number of tails. List the four outcomes and the value of X for each.

3. *(Understand)* A health card records heights to the nearest centimetre. Explain, with one example, why height is still a continuous random variable.

4. *(Analyze)* A trainee says: "Marks on a 20-mark test are continuous, because marks measure ability and ability can take any value." Analyze this claim: separate what is measured from what is recorded, and say whether the random variable "marks" is discrete or continuous.

5. *(Apply)* Classify each as discrete or continuous and give a one-line reason: (a) the number of tiffin boxes in a classroom, (b) rainfall in Karachi on a July day in mm, (c) the number of goals in a school football match, (d) the time a pupil spends on homework in an evening.

6. *(Apply)* Over 40 days a teacher recorded the number of pupils who arrived late: 0 late on 8 days, 1 on 16 days, 2 on 12 days, 3 on 4 days. Build the probability distribution and run both checks.

7. *(Apply)* Compute the expected number of late pupils per day from your distribution in RRQ 6 and state what it means in one sentence.

8. *(Analyze)* A trainee gives $P(X = 0) = 0.25$, $P(X = 1) = 0.25$, $P(X = 2) = 0.25$, $P(X = 3) = 0.30$ for the number of pupils without a textbook. Analyze whether this is a valid distribution and suggest what mistake might have produced it.

9. *(Understand)* For the Larkana absentees, the expected value is 2.12 and the most likely value is 2. Explain the difference between these two figures.

10. *(Analyze)* A teacher says: "The score on a fair die is a random variable and every score is equally likely, so every random variable has equally likely values." Analyze this claim using the number of heads when two coins are tossed.

Extended-response questions (ERQs)

Answer each question in a structured paragraph or set of paragraphs.

1. *(Analyze)* A district education officer's form asks each school for: (a) the number of teachers, (b) the area of the playground in square metres, (c) the monthly electricity bill in rupees, (d) the number of pupils who passed Class 5, (e) the average daily temperature in the staffroom, and (f) the marks of the top pupil out of 100. Name each random variable, describe its possible values and classify it. Then analyze the two cases most likely to cause disagreement and justify your decision. (150-200 words after your table.)

2. *(Evaluate)* Two teachers propose distributions for X, the number of pupils without homework in a class on a given day.

x	0	1	2	3
Teacher A	0.30	0.45	0.35	-0.10
Teacher B	0.25	0.40	0.25	0.10

Evaluate each table against both checks, decide which can be used, and compute the expected value of the valid one. Explain why Teacher A's table would mislead a head teacher even though its probabilities add to 1. (150-200 words after your working.)

3. *(Analyze)* At a school mela in Sukkur, a stall charges Rs 20 to roll one fair die. A player who rolls a 6 wins Rs 60; any other score wins nothing. Let X be the amount won. Build the distribution of X, compute $E(X)$, and analyze whether the game is fair to players. Estimate what the school should expect to earn from 300 plays, and explain why the actual earnings on the day may differ. (150-200 words after your working.)

4. *(Evaluate)* Using the Larkana absentee distribution (0: 0.10, 1: 0.24, 2: 0.30, 3: 0.20, 4: 0.12, 5: 0.04), a head teacher decides to print exactly 2 substitute worksheets each day "because 2 absentees is what we expect". Evaluate this decision. Compute the expected value and the probability that more than 2 pupils are absent, and recommend a better plan with reasons. (150-200 words after your working.)

5. *(Evaluate)* A trainee plans a practicum study: "I will record the number of absentees and each pupil's height for 20 days. Both are discrete because I will write them as whole numbers. I will make a probability table for each by listing every value once and giving it the same probability, and the expected value will tell me the number of absentees on most days." Evaluate the plan across both topics: identify each error, explain why it is an error, and rewrite the plan so it would produce valid distributions and a correct interpretation. (200-250 words.) Then ask your cooperating teacher for five actual daily absentee totals from one class, with no names or individual records.

Build and check their relative-frequency table and calculate its expected value. Name the district and collection period, compare the real data with the invented Larkana example, and explain why five days cannot establish future probabilities. If school access is unavailable, use five days of your own B.Ed group attendance with consent; identify the source and keep names out of your response.

Answers and marking guidance

MCQ answer key

Item	Answer	Justification
1	A	A random variable attaches a number to each outcome of a chance process.
2	B	Pupils wearing glasses are counted: separate whole numbers.
3	C	Arm length is measured and can take any value in an interval.
4	C	Three coins can show 0, 1, 2 or 3 heads.
5	B	Age can take any value between whole years; recording rounds it.
6	B	The two checks: each probability in 0-1 and a total of 1.
7	B	1 - (0.2 + 0.3 + 0.1) = 0.4.
8	A	0 × 0.5 + 1 × 0.3 + 2 × 0.2 = 0.7.

Item	Answer	Justification
9	B	The expected value is a long-run average.
10	B	For a continuous variable, probability is area under the curve.

RRQ model answers and mark schemes

1. The number of heads is a number attached to each outcome (HH gives 2, HT and TH give 1, TT gives 0). It is random because which value we get is decided by chance before the toss. (2 marks: 1 for a number per outcome, 1 for chance deciding the value.)

2. HH: 0, HT: 1, TH: 1, TT: 2. (2 marks: 1 for the four outcomes, 1 for all values right.)

3. Height can take any value in an interval: a pupil recorded as 139 cm might be 138.6 cm or 139.4 cm. Rounding is how we record it, not a property of the height itself. (2 marks: 1 for any value in between, 1 for rounding being recording only.)

4. Ability may vary continuously, but the random variable here is the mark recorded, and marks are awarded in separate steps (whole or half marks) that can be listed and counted. So "marks out of 20" is discrete, even if it is used to estimate something continuous. (3 marks: 1 for separating measured from recorded, 1 for separate listable values, 1 for the verdict discrete.)

5. (a) discrete, counted; (b) continuous, measured and can take any value; (c) discrete, counted; (d) continuous, time is measured. (2 marks: 4 correct with reasons = 2, 2-3 correct = 1.)

6. Probabilities 8/40 = 0.20, 16/40 = 0.40, 12/40 = 0.30, 4/40 = 0.10. Each lies between 0 and 1, and they add to 1.00, so the table is valid.

(3 marks: 2 for the four probabilities, 1 for both checks stated.)

7. $E(X) = 0 \times 0.20 + 1 \times 0.40 + 2 \times 0.30 + 3 \times 0.10 = 1.30$. Over many days, about 1.3 pupils arrive late per day on average. (3 marks: 1 for the method, 1 for 1.30, 1 for the long-run meaning.)

8. Each value lies between 0 and 1, but the total is 1.05, so it fails the second check and is not valid. A likely cause is counting some days twice, or rounding each probability up. The trainee should recount the frequencies and divide each by the same total. (3 marks: 1 for check 1 passed, 1 for check 2 failed with the sum, 1 for a plausible cause.)

9. The most likely value, 2, is the single value with the highest probability (0.30), the mode of the distribution. The expected value, 2.12, is the long-run average over many days, weighting every value by its probability; it is not a value any one day can have. (2 marks: 1 for each figure explained.)

10. The claim is false. For two coins, $X = 0, 1, 2$ have probabilities 0.25, 0.50 and 0.25, so 1 head is twice as likely as 0 heads. Each random variable has its own distribution; the fair die is a special case where values are equally likely. (3 marks: 1 for rejecting the claim, 1 for the two-coin distribution, 1 for naming the die as a special case.)

ERQ rubrics

ERQ 1 (Analyze - classifying a district form)

Expected classification: (a) discrete, (b) continuous, (c) discrete in paisa or rupees, though often treated as continuous because the steps are so small, (d) discrete, (e) continuous, (f) discrete if whole marks. Accept either verdict on (c) when it is reasoned.

Criterion	Band 3 (Full)	Band 2 (Partial)	Band 1 (Limited)	Band 0
Variables and values	All six named with sensible possible values	Four or five	Two or three	None
Classification	All six defensible	Four or five	Two or three	None
Analysis of hard cases (Analyze)	Picks two genuinely ambiguous cases, applies a named test, and separates the quantity from how it is recorded	One case well argued	Verdicts without reasons	None

ERQ 2 (Evaluate - judging two proposed distributions)

Correct values: Teacher A sums to 1.00 but has -0.10, so it fails check 1. Teacher B passes both; $E(X) = 0 + 0.40 + 0.50 + 0.30 = 1.20$.

Criterion	Band 3 (Full)	Band 2 (Partial)	Band 1 (Limited)	Band 0
Checks applied	Both checks on both tables, verdicts right	One table right	Checks named, misapplied	None
Expected value	1.20 with working	Method right, slip	Wrong method	None

Criterion	Band 3 (Full)	Band 2 (Partial)	Band 1 (Limited)	Band 0
Evaluation (Evaluate)	Explains that a negative probability is impossible, that the other values are then too large, and that calculations from invalid probabilities cannot support a planning			
judgement; judges B usable with a reason	Judgement with partial reasoning	Verdict only	None	

ERQ 3 (Analyze - expected value of a mela game)

Correct values: $X = 60$ with probability 1/6 and 0 with probability 5/6; $E(X) = 10$. A player pays Rs 20, so expects to lose Rs 10 per play; 300 plays give the school about 300 × 10 = Rs 3,000.

Criterion	Band 3 (Full)	Band 2 (Partial)	Band 1 (Limited)	Band 0
Distribution and $E(X)$	Both correct	One correct	Attempted	None
Fairness and earnings	Compares Rs 10 with the Rs 20 fee and	One of the two	Unsupported claim	None

Criterion	Band 3 (Full)	Band 2 (Partial)	Band 1 (Limited)	Band 0
	estimates Rs 3,000			
Analysis of variation (Analyze)	Explains that Rs 3,000 is a long-run expectation and that the real day may differ because 6s occur by chance	Mentions chance without linking to the estimate	Treats the estimate as certain	None

ERQ 4 (Evaluate - planning from an expected value)

Correct values: $E(X) = 2.12$; $P(X > 2) = 0.20 + 0.12 + 0.04 = 0.36$.

Criterion	Band 3 (Full)	Band 2 (Partial)	Band 1 (Limited)	Band 0
Computation	Both values correct	One correct	Attempted	None
Evaluation of the decision (Evaluate)	Shows the head teacher has mixed up the most likely value with the expected value, and that on about 36% of days 2	Notes the shortfall without the probability or the mix-up	Agrees or disagrees without evidence	None

Criterion	Band 3 (Full)	Band 2 (Partial)	Band 1 (Limited)	Band 0
	worksheets will not be enough			
Recommendation	A justified plan (for example print 4, covering 0.96 of days, or keep a small stock based on 2.12 per day over a week)	A plan with weak reasons	A plan with no reason	None

ERQ 5 (Evaluate - integrative: judging a study plan)

Errors to find: height is continuous, not discrete; a distribution built from records must use relative frequencies, not equal probabilities; height needs intervals or a curve, not one probability per value; the expected value is a long-run average, not the value on most days.

Criterion	Band 3 (Full)	Band 2 (Partial)	Band 1 (Limited)	Band 0
Errors identified	All four	Three	One or two	None
Explanation (Evaluate)	Each error explained with the correct	Most explained	Errors listed without reasons	None

Criterion	Band 3 (Full)	Band 2 (Partial)	Band 1 (Limited)	Band 0
	principle from Topic 10.1 or 10.2			
Rewritten plan	Classifies correctly, builds distributions from frequencies, runs both checks, groups heights into intervals and interprets $E(X)$ as a long-run average	Mostly correct	Partly corrected	None

Criterion	Band 3 (Full)	Band 2 (Partial)	Band 1 (Limited)	Band 0
Real-data application (Analyze)	Gives five anonymous totals with district and period, a valid table and mean, and a justified comparison with the example including the small-sample limit	Data and calculation with limited comparison	Data without interpretation or provenance	None

Was this page helpful? Yes

Tip: select any text on this page first to comment on that specific passage - or leave nothing selected for general feedback.

Give feedback